

Part 01 : Theoretical Questions

Q1 : Identify the type of relationship in each scenario below (Inheritance, Association, Aggregation, Composition, or Dependency):

- a) A University has Departments. If the university is closed, the departments no longer exist.
- b) A Driver uses a Car. The driver does not own the car.
- c) A Dog is an Animal.
- d) A Team has Players. If the team is deleted, the players still exist.
- e) A method receives a Logger as a parameter and calls it inside the method only.

Q2 : Answer the following questions about access modifiers and sealed:

- a) A parent class has a protected field. Can a child class in a different assembly access it? What about through an object instance from outside?
- b) What is the difference between protected internal and private protected?
- c) What does the sealed keyword do when applied to a class? What about when applied to a method?

d) Can you create an object from a sealed class using new? Why or why not?

Part 02 : Practical (Extending the Movie Ticket Booking System)

In Assignments 01 & 02, you built a Movie Ticket Booking System with a Ticket class, properties, indexers, and static members. Now you will refactor and extend it using inheritance, composition, and sealed classes.

What you need to build :

1. Create a base class Ticket with:
 - a. MovieName (string), Price (decimal, must be > 0), TicketId (int, read-only, auto-incremented).
 - b. A constructor that takes movieName and price.
 - c. A computed property PriceAfterTax that returns the price with 14% tax.
 - d. Override ToString() to return the ticket info.
 - e. A static int GetTotalTickets() method that returns the total number of tickets created.
2. Create three child classes that inherit from Ticket:
 - a. StandardTicket — adds SeatNumber (string).
 - b. VIPTicket — adds LoungeAccess (bool) and ServiceFee (decimal) = 50.
 - c. IMAXTicket — adds Is3D (bool). If true, the price increases by 30 EGP.Each child class should override ToString() to include its own extra info.

3. Create a Cinema class that has a CinemaName, a Projector object (created inside Cinema), and holds up to 20 tickets. Add:

- a. AddTicket(Ticket t) — adds a ticket to the first available slot.
- b. PrintAllTickets() — prints all tickets.
- c. OpenCinema() and CloseCinema() — start/stop the projector.

4. In Main, do the following:

- a. Create a Cinema and open it.
- b. Create one of each ticket type (hardcoded data) and add them to the Cinema.
- c. Print all tickets.
- d. Close the Cinema.

Expected Output (Example) :

```
===== Cinema Opened =====
Projector started.

===== All Tickets =====
Ticket #1 | Inception | Price: 120 EGP | After Tax: 136.80 EGP |
Seat: A-5
Ticket #2 | Avengers | Price: 200 EGP | After Tax: 228.00 EGP |
Lounge: Yes | Service Fee: 50 EGP
Ticket #3 | Dune | Price: 180 EGP | After Tax: 205.20 EGP | IMAX
3D: No

===== Statistics =====
Total Tickets Created: 3

Booking Ref 1: BK-1
Booking Ref 2: BK-2
```